Meeting to Action

System Design Document

Status: Advanced single-meeting branch publicly deployed and verified **Last updated:** October 14, 2026 **Branch:** `feature/advanced-single-meeting` (`5566b02`); `main` remains separate **Primary runtime:** Python 3.12 and Streamlit **AI platform:** Microsoft Foundry with Microsoft Agent Framework

1. Executive Summary

Meeting to Action converts one meeting transcript into a structured, evidence-grounded record of notes, actions, decisions, and unresolved questions. A reviewer can correct and approve that record, ask cited questions about the meeting, and prepare follow-up calendar events. The system is intentionally scoped to one meeting at a time and does not autonomously execute business actions.

The current implementation is a modular monolith: Streamlit owns the user interface and session workflow, while focused Python modules own ingestion, extraction, retrieval, grounding, human review, connector access, calendar preparation, and evaluation. Microsoft Foundry hosts the generation and independent judge deployments. Pydantic contracts and deterministic evidence checks form the boundary between probabilistic model output and trusted application state.

The central design principle is that fluent model output is not proof. Every accepted record must carry an exact transcript quote and line range, survive deterministic validation, and pass human review before it can support question answering or a downstream calendar action.

The advanced branch adds Azure Speech recording/live-clip ingestion, timestamped speaker segments, supporting-document alignment, explainable conflict checks, actor-attributed correction history, an evidence-linked playback timeline, and approval-gated Outlook/Teams workflow previews. It remains scoped to one meeting in session memory. This branch is deployed publicly, but normal traffic still uses `gpt-5.4-mini`; the experimental fine-tuned model is isolated to Model Lab.

2. Problem Definition

2.1 Problem specificity

Meeting transcripts contain commitments and decisions that are difficult to retrieve and convert into reliable follow-up records. General summarization is insufficient because it can omit important outcomes, infer owners or dates, and lose traceability to the source.

The system solves a narrow problem:

> Given one meeting transcript, produce a typed, evidence-grounded meeting record that a person > can review, approve, query, and use to prepare calendar follow-up.

2.2 In scope

- Import one transcript from a local file or configured remote source.
- Normalize the source into stable, numbered lines.
- Extract notes, actions, decisions, and open questions.

- Compare three model orchestration strategies.
- Validate evidence quotes, line ranges, and explicit deadline years.
- Let a human edit and approve the structured record.
- Record field-level changes made during review.
- Answer meeting questions from retrieved evidence and the approved record.
- Prepare calendar drafts only after explicit user confirmation.
- Transcribe uploaded recordings and successive microphone clips through an existing Azure Speech resource.
- Compare transcript outcomes with supporting documents without treating those documents as occurrence evidence.
- Explain missing fields and potential conflicts without unsupported confidence percentages.
- Prepare Outlook drafts and Teams previews only after explicit user approval.
- Evaluate quality, grounding, latency, and model-call cost.

2.3 Out of scope

- Autonomous task execution or automatic invitation delivery.
- Organization-wide knowledge search.
- Cross-meeting action tracking.
- Durable multi-user workflow and reviewer identity management.
- Automatic PII redaction from transcript content.
- Using the LLM judge as the sole quality or safety control.

3. Requirements

3.1 Functional requirements

ID	Requirement
FR-1	Accept pasted text and TXT, VTT, SRT, DOCX, or EML files up to 10 MB.
FR-2	Read transcripts from configured Microsoft 365, Teams, Azure Blob, Google Drive, Gmail, and Zoom sources.
FR-3	Normalize content into stable numbered lines while preserving speaker text.
FR-4	Produce a schema-valid `MeetingExtraction`.
FR-5	Attach at least one exact evidence citation to every extracted record.
FR-6	Support single-model, extractor-verifier, and specialist-consolidator strategies.
FR-7	Allow human editing while preserving source evidence and recording changed fields.
FR-8	Answer supported questions with validated citations and reject unsupported topics.

FR-9	Require explicit approval before creating or opening a calendar draft.
FR-10	Evaluate all model strategies on a frozen benchmark with deterministic and judged metrics.
FR-11	Transcribe audio/video with speaker, locale, and timestamp metadata through Azure Speech.
FR-12	Build an evidence-linked timeline and seek recording playback to known timestamps.
FR-13	Record reviewer-attributed, timestamped field corrections while preserving evidence.
FR-14	Keep supporting documents separate from authoritative transcript evidence.
FR-15	Gate Outlook drafts, Teams deep links, and Adaptive Card downloads behind approval.

3.2 Quality attributes

Attribute	Design response
Grounding	Exact quote and line-range validation after every extraction and answer.
Safety	No inferred owners or years; no autonomous calendar action; human approval gate.
Explainability	Every record exposes source evidence; review changes are auditable.
Maintainability	Domain behavior is separated into small Python modules with typed contracts.
Testability	External/model boundaries are mockable; 70 automated tests cover core and advanced behavior.
Cost awareness	Architecture comparison records generation calls and latency per strategy.
Privacy	Credentials and connector metadata are excluded from prompts; transcript content is sent only to the configured Foundry model.
Availability	Azure Container Apps scales from zero to one replica; there is no multi-replica or cross-region guarantee.

4. System Context

Architecture flow

```
flowchart LR
    User[Meeting organizer or reviewer]
    App[Meeting to Action\nStreamlit application]
    Sources[Transcript sources\nFiles, M365, Teams, Blob,\nGoogle, Gmail, Zoom]
    Speech[Azure Speech\nFast transcription]
    Context[Supporting documents\nTXT, DOCX, PDF]
    Foundry[Microsoft Foundry\nGeneration deployment]
    Judge[Microsoft Foundry\nIndependent judge deployment]
    Calendars[Calendar providers\nGoogle, Outlook, Yahoo, ICS]
    Workflows[Outlook and Teams\nDraft destinations]

    User -->|Import, review, approve, ask| App
    Sources -->|Read-only transcript content| App
    Speech -->|Diarized text and timestamps| App
    Context -->|Non-authoritative alignment context| App
    App -->|Normalized transcript and typed prompts| Foundry
    Foundry -->|Structured extraction or answer| App
    App -->|Benchmark reference and candidate| Judge
    App -->|Approved event draft| Calendars
    App -->|Approved message or card draft| Workflows
```

4.1 Actors

- **Meeting organizer:** imports a transcript and selects an extraction strategy.
- **Reviewer:** verifies evidence, edits fields, approves the record, and authorizes calendar work.
- **Calibration reviewer:** independently scores blinded evaluation candidates.
- **Cloud administrator:** provisions Foundry resources, deployments, access roles, and monitoring.

4.2 External dependencies

- Microsoft Foundry project and model deployments.
- Existing Azure Speech endpoint with fast transcription enabled.
- Azure CLI or managed identity credentials for Foundry and Azure Blob access.
- Microsoft Graph delegated authentication for Microsoft 365 and Teams.
- Google OAuth for Drive, Gmail, and Calendar.
- Zoom Server-to-Server OAuth for cloud transcripts.
- Browser or desktop calendar handlers for provider drafts and ICS files.

5. Logical Architecture

Architecture flow

```
flowchart TB
    UI[Streamlit UI and session workflow]
    Connectors[Connector adapters]
    Ingestion[Ingestion and normalization]
    Speech[Speech transcription]
    Context[Context parsing]
    Transcript[Numbered transcript]
    Extraction[Extraction orchestration]
    Contracts[Pydantic domain contracts]
    Validation[Deterministic grounding validation]
    Review[Human review and change audit]
    Quality[Quality and conflict analysis]
    Timeline[Evidence timeline]
    Retrieval[BM25 line retrieval]
```

```
Ask[Grounded question answering]
Calendar[Calendar draft and approval gate]
Workflows[Outlook and Teams drafts]
ModelLab[Single-pass Model Lab]
Evaluation[Benchmark and calibration]
Foundry[FoundryChatClient]

UI --> Connectors
UI --> Ingestion
UI --> Speech --> Ingestion
UI --> Context --> Quality
Connectors --> Ingestion
Ingestion --> Transcript
Transcript --> Extraction
Extraction --> Foundry
Foundry --> Contracts
Contracts --> Validation
Validation --> Review
Validation --> Quality
Validation --> Timeline
Review --> UI
Review --> Retrieval
Transcript --> Retrieval
Retrieval --> Ask
Ask --> Foundry
Ask --> Validation
Review --> Calendar
Review --> Workflows
Validation --> ModelLab
Evaluation --> Extraction
Evaluation --> Foundry
```

5.1 Component responsibilities

Component	Responsibility
app.py	UI composition, session-state workflow, user events, review tabs, and action gating.
meeting_to_action/connectors.py	Read-only authentication and retrieval adapters for remote transcript sources.
meeting_to_action/ingestion.py	File validation, format parsing, metadata extraction, normalization, and content hashing.
meeting_to_action/transcript.py	Stable line numbering, quote lookup, grounding validation, and repair.
meeting_to_action/models.py	Shared typed contracts for evidence, notes, actions, decisions, questions, and complete records.
meeting_to_action/extraction.py	Foundry client construction and the three extraction orchestration strategies.
meeting_to_action/retrieval.py	BM25 ranking and neighboring-line context expansion.
meeting_to_action/chat.py	Grounded factual answers and labeled analytical synthesis.
meeting_to_action/review.py	Reconstruction of reviewed records while preserving evidence and producing field-level audits.

meeting_to_action/calendar.py	Calendar payload generation, provider dispatch, and approval enforcement.
meeting_to_action/advanced.py	Speech segments, context citations, quality findings, corrections, and timeline contracts.
meeting_to_action/speech.py	Azure Speech fast transcription and phrase-to-segment normalization.
meeting_to_action/quality.py	Deterministic missing-field, context-alignment, and conflict explanations.
meeting_to_action/timeline.py	Maps transcript evidence lines to speech timestamps and speakers.
meeting_to_action/workflows.py	Outlook/Teams preview payloads and Adaptive Card generation.
meeting_to_action/advanced_evaluation.py	WER, timestamp, diarization, conflict, and live/batch metrics.
meeting_to_action/evaluation.py	Frozen benchmark execution, deterministic metrics, semantic F1, and independent judge calls.
meeting_to_action/calibration.py	Blinded reviewer packets and judge-human agreement calculation.

6. Data Design

6.1 Primary data objects

`TranscriptDocument` contains source type, external identifier, title, filename, normalized text, SHA-256 content hash, import timestamp, MIME type, and source metadata.

`MeetingExtraction` is the aggregate root for generated and approved meeting records:

Architecture flow

```
classDiagram
    class MeetingExtraction {
        string title
        date meeting_date
        string summary
        MeetingNote[] notes
        ActionItem[] actions
        Decision[] decisions
        OpenQuestion[] open_questions
        string[] warnings
    }
    class Evidence {
        string quote
        int line_start
        int line_end
    }
    class ActionItem {
        string id
        string task
        string owner
```

```
    string deadline_text
    date deadline
    string status
}
class Decision {
    string id
    string summary
    string rationale
}
class OpenQuestion {
    string id
    string question
    string owner
}
class MeetingNote {
    string id
    string title
    string description
}

MeetingExtraction "1" *-- "many" MeetingNote
MeetingExtraction "1" *-- "many" ActionItem
MeetingExtraction "1" *-- "many" Decision
MeetingExtraction "1" *-- "many" OpenQuestion
MeetingNote "1" *-- "one or more" Evidence
ActionItem "1" *-- "one or more" Evidence
Decision "1" *-- "one or more" Evidence
OpenQuestion "1" *-- "one or more" Evidence
```

6.2 State and persistence

The current application stores imported content, generated records, approved records, chat history, selected evidence, and change audit in Streamlit session state. Evaluation datasets and results are local JSON or CSV files. OAuth token files are local and ignored by source control.

Recording bytes, speech segments, parsed supporting documents, quality findings, timeline state, and correction events also remain in the active Streamlit session. Supporting documents are not inserted into transcript evidence. A new import or extraction invalidates downstream reviewed state; there is no cross-meeting database.

This design is suitable for a single-user capstone but is not a durable system of record. A production design should persist approved records and immutable audits in an authenticated data store, encrypt data at rest, associate changes with reviewer identity, and apply retention rules.

6.3 Data lifecycle

1. Import raw bytes or remote text.
2. Validate size and supported format.
3. Parse and normalize text.
4. Compute a content hash and clear stale derived state when the source changes.
5. Number transcript lines and submit only required content to Foundry.
6. Validate and repair model evidence.
7. Store generated and reviewed objects in the active session.
8. Export audits or calendar payloads only at user request.
9. Lose session data when the process or session ends unless the user exported it.

For recording upload, raw media bytes remain in the active session to support timestamp seeking. For live mode, each submitted browser microphone clip is transcribed and appended with adjusted segment offsets;

clips are not retained as one continuous playable recording. Supporting document text and correction events also remain session-scoped and are cleared with the meeting workflow.

7. Core Processing Flows

7.1 Transcript ingestion and extraction

Architecture flow

```
sequenceDiagram
    actor U as User
    participant UI as Streamlit UI
    participant I as Ingestion
    participant E as Extraction strategy
    participant F as Foundry model
    participant V as Grounding validator

    U->>UI: Select source and import transcript
    UI->>I: Bytes or remote transcript
    I->>I: Validate, parse, normalize, hash
    I-->>UI: TranscriptDocument
    U->>UI: Select strategy and extract
    UI->>E: Numbered transcript and title
    E->>F: Typed extraction prompt
    F-->>E: MeetingExtraction candidate
    E->>V: Candidate and numbered transcript
    V->>V: Validate quotes, ranges, and deadlines
    V-->>UI: Grounded record plus warnings
```

7.2 Human review and approval

The reviewer edits business fields but cannot silently replace model evidence through the editable table. `apply_review_edits` reconstructs every record using the original evidence. Pydantic revalidates the result. `build_change_audit` compares audited fields and emits before/after values. Only the approved record is used by grounded Ask and calendar preparation.

7.3 Grounded question answering

10. BM25 ranks normalized transcript lines for the question.
11. The top eight lines are expanded by one line on either side.
12. Original line numbers are retained.
13. Retrieved evidence and the approved meeting record are sent to the answer agent.
14. A supported answer must contain at least one citation.
15. Every citation is validated against the complete transcript.
16. Invalid citations fail the request rather than reaching the UI as trusted output.

7.4 Calendar action

The calendar module receives only an approved action and user-entered scheduling details. The UI shows the complete event payload and requires explicit confirmation. Google Calendar uses OAuth; Outlook and Yahoo open user-reviewed drafts; other providers use an ICS download. The system does not automatically send invitations.

7.5 Recording and live-clip transcription

17. The user selects **Upload audio/video recording** or **Record live meeting clips**.
18. The UI validates a supported AAC, FLAC, M4A, MOV, MP3, MP4, OGG, WAV, WEBM, or WMA payload of

at most 300 MB and requires a speaker maximum between 2 and 35.

19. `speech.py` calls Azure Speech fast transcription with candidate locales and diarization.
20. Recognized phrases become immutable `TranscriptSegment` records with text, speaker, locale, start milliseconds, and end milliseconds.
21. The normalized speaker-labelled text becomes the authoritative transcript. Uploaded media is retained for timeline playback; live segments are appended with monotonic offsets.

7.6 Quality, context, and timeline

`quality.py` deterministically flags missing owners and deadlines. It reports context alignment when action-token overlap with a supporting-document line reaches `0.4`. It reports a potential conflict when similar action text has Jaccard overlap of at least `0.65` and explicit owners or deadlines differ. Context citations do not become transcript evidence. `timeline.py` maps the first evidence range for each action, decision, and question to the corresponding Speech segment when available and otherwise falls back to line order.

7.7 Correction and communication workflow

Approval compares generated and reviewed records and appends actor-attributed `CorrectionEvent` entries containing timestamp, record type, record ID, field, and before/after values. An approved action can then produce Outlook and Teams deep links or an Adaptive Card 1.5 download. With a delegated Microsoft token, Graph may create a saved `/me/messages` draft. There is no automatic send endpoint.

7.8 Model Lab

Model Lab compares exactly one production single-model extraction with one trained-model single-model extraction on the same transcript. It reuses the main result only when the selected strategy was already single-model; otherwise it requests a separate production pass. The result is session-scoped and has no promotion or training-data side effect.

8. Model Orchestration

8.1 Strategies

Strategy	Calls	Design	Primary tradeoff
Single model	1	One agent returns the complete record.	Lowest latency and cost; no semantic verification stage.
Extractor + verifier	2	A second agent audits and corrects the first result.	Better separation of generation and verification at moderate cost.
Specialists + consolidator	4	Three concurrent specialists feed one consolidator.	Highest decomposition and model cost; useful for dense meetings.

All strategies share the same typed contracts and deterministic validation. The selected default is extractor + verifier because it provides an explicit audit boundary, achieved the highest semantic F1 in the frozen run, and

uses half the generation calls of the specialist design. This is a tradeoff decision, not a claim of dominance across every metric.

8.2 Context handling

- Extraction receives the complete numbered transcript for one meeting.
- Ask receives a BM25-selected evidence window plus the approved structured record.
- Prompts prohibit outside facts and inferred owners or deadlines.
- A normalized date is allowed only when the cited transcript explicitly states the year.
- Provider credentials, tokens, URLs, and connector metadata are not included in prompts.

8.3 Model deployments

Role	Deployment	Model/version	Tier and capacity	Routing
Production generation	`gpt-5.4-mini`	`gpt-5.4-mini`, `2026-03-17`	GlobalStandard, 10	`FOUNDRY_MODEL`; default extraction and Ask
Independent judge	`gpt-4.1-judge`	`gpt-4.1`, `2025-04-14`	GlobalStandard, 10	`FOUNDRY_JUDGE_MODEL`; offline evaluation only
Experimental generation	`meeting-to-action-sft-v1`	Fine-tuned `gpt-4.1-mini`, `2025-04-14`, checkpoint step 48	DeveloperTier, 50	`FOUNDRY_TRAINED_MODEL`; Model Lab only

The deployments are in East US 2 under Foundry account `ai-account-nlmcnaciofa6y` and project `ai-project-meetingtoaction`. Model Lab forces a one-pass comparison on both deployments. It reuses an existing production result only when that result came from the single-model strategy and invokes the trained deployment through an explicit model override. It does not mutate `FOUNDRY_MODEL`, write training data, or affect the normal extraction, review, Ask, or calendar paths.

- The evaluation code rejects a missing judge deployment or one with the same name as the generation deployment.

9. Retrieval Design

BM25 is used instead of a vector database because the active search corpus is one transcript, questions frequently contain exact names or deadlines, and line-level auditability is mandatory. Neighbor expansion preserves local conversational context without sending the full transcript for every question.

Advantages:

- Deterministic and inspectable ranking.
- No embedding service or persistent index required at runtime.
- Original evidence line numbers remain stable.
- Low operational complexity for a single document.

Limitations:

- Lexical retrieval can miss paraphrases.
- It does not support organization-wide or cross-meeting retrieval.
- There is no reranker or learned relevance model.

A production cross-meeting design should add tenant-filtered hybrid retrieval, document-level authorization filters, index deletion on retention events, and retrieval quality evaluation before changing the current implementation.

10. Security, Privacy, and Guardrails

10.1 Trust boundaries

Architecture flow

```
flowchart LR
    subgraph UserDevice[User device and application process]
        UI[Streamlit]
        State[Session state]
        Guard[Typed contracts and deterministic validation]
    end
    subgraph Azure[Azure trust boundary]
        Project[Foundry project]
        Models[Model deployments]
    end
    subgraph Providers[External provider boundaries]
        Graph[Microsoft Graph]
        Google[Google APIs]
        Zoom[Zoom API]
    end

    Providers -->|Transcript content| UI
    UI --> Guard
    Guard -->|Required prompt content| Models
    Models --> Guard
    Guard --> State
```

10.2 Identity and access

- Foundry uses `AzureCliCredential` locally and `ManagedIdentityCredential` in the deployed

Container App.

- The Container App system identity has `Foundry User` at project scope and `AcrPull` at registry

scope, plus `Cognitive Services Speech User` at AI account scope; none of these assignments is subscription- or resource-group-scoped.

- Microsoft Graph uses delegated permissions so access follows the signed-in user.
- Azure Blob access requires the Storage Blob Data Reader role.
- Google and Zoom use provider-specific OAuth credentials and least-privilege read scopes.
- Infrastructure assigns Azure roles to an explicit user or application principal.

10.3 PII handling

Meeting transcripts may contain names, email addresses, customer details, or other personal data. The current system does not automatically detect or redact PII. Transcript content is processed in application memory and sent to the configured Foundry model for extraction or question answering. Credentials, access tokens, source URLs, and connector metadata are excluded from model prompts.

Production requirements should include documented user consent, tenant-approved model data handling, data classification, configurable redaction, regional processing requirements, retention and deletion policies, encrypted durable storage, and access/audit controls.

10.4 Guardrails

- Pydantic rejects malformed or incomplete structured output.
- Evidence is mandatory for every extracted record.
- Exact quotes must occur within declared transcript ranges.
- Unsupported records are removed or surfaced through warnings.
- Owners and deadlines cannot be inferred when not explicit.
- Supported answers require validated citations.
- Unsupported questions receive an explicit not-covered response.
- Human approval gates all downstream calendar actions.
- Independent human calibration constrains interpretation of judge scores.

11. Deployment Architecture

11.1 Current environment

The application is deployed and publicly reachable at https://ca-mta-nlmcnaci.greenground-3fac55c9.eastus2.azurecontainerapps.io/. The hosted runtime uses Azure Container Apps in East US 2 and references the existing Microsoft Foundry account and project without modifying them. The generation and independent judge deployments remain separate; only `gpt-5.4-mini` is used by normal extraction and Ask, while `meeting-to-action-sft-v1` is available only through the explicit Model Lab control.

Architecture flow

```
flowchart TB
        Browser[User browser]
        Ingress[Public HTTPS ingress\nport 8501]
        Streamlit[Azure Container App\nca-mta-nlmcnaci]
        Identity[System-assigned managed identity]
        ACR[Authenticated Azure Container Registry\ncrmtanlmcnaciof]
    FoundryProject[Microsoft Foundry project]
    Generation[gpt-5.4-mini]
    Trained[meeting-to-action-sft-v1\nModel Lab only]
    Speech[Azure Speech endpoint]
        Logs[Existing Log Analytics workspace]

        Browser --> Ingress --> Streamlit
        ACR -->|Application image| Streamlit
        Streamlit --> Identity
        Identity -->|Foundry User| FoundryProject
    FoundryProject --> Generation
    FoundryProject --> Trained
        Identity -->|Speech User| Speech
        Streamlit -. Runtime logs .-> Logs
```

Deployment property	Live value
Region	East US 2
Container App	`ca-mta-nlmcnaci`
Container Apps environment	`cae-mta-nlmcnaci`
Registry	`crmtanlmcnaciof`, Basic, admin user disabled
Runtime	Python 3.12 non-root image, single revision, 0-1 replicas

Ingress	Public HTTPS only, insecure transport disabled
Health	Root and `/_stcore/health` return HTTP 200
Identity	System-assigned; project-scoped `Foundry User`, registry-scoped `AcrPull`, account-scoped `Cognitive Services Speech User`
Live AI verification	Extractor + verifier returned grounded actions, decisions, notes, and questions
Active revision	`ca-mta-nlmcnaci--azd-1791970708`
Active image	`crmtanlmcnaciof.azurecr.io/ai-foundry-starter-basic/web-meetingtoaction:azd-deploy-1791970619`
Speech	Account endpoint configured without an API key; managed-identity synthetic WAV transcription verified
Model Lab	Step-48 deployment `Succeeded`; direct grounded single-pass inference verified; production routing unchanged

Local development remains available at `localhost:8501` and uses `AzureCliCredential`. Hosted Google, Zoom, and delegated Microsoft connector flows are not enabled until server-safe OAuth callbacks and durable token storage are designed.

The deployed environment sets `FOUNDRY_PROJECT_ENDPOINT`, `FOUNDRY_MODEL=gpt-5.4-mini`, `FOUNDRY_TRAINED_MODEL=meeting-to-action-sft-v1`, and the account-scoped `AZURE_SPEECH_ENDPOINT`. It does not set `AZURE_SPEECH_API_KEY`. Local development can use the same endpoint with Azure CLI RBAC or an optional key that remains outside source control.

11.2 Production hardening roadmap

- Add Microsoft Entra authentication and per-user authorization; the current endpoint is

intentionally public.

- Store secrets in Key Vault and never in application files.
- Persist approved records and immutable audits in a tenant-partitioned database.
- Enable Application Insights with privacy-reviewed telemetry.
- Add health probes, deployment slots or revisions, backups, and recovery objectives.
- Keep generation and judge deployments logically separate.
- Apply network controls and private endpoints where organizational policy requires them.

12. Reliability and Failure Handling

Failure	Current response
Unsupported file or file over 10 MB	Reject before parsing.
Empty or unreadable transcript	Reject with a validation error.
Connector authentication/API failure	Surface a connector-specific error without sending

	credentials to the model.
Malformed model response	Fail Pydantic validation.
Unsupported evidence quote	Remove or reject the affected record and emit a warning.
Incorrect line range with uniquely found quote	Repair the range deterministically.
Invalid answer citation	Fail the grounded answer.
Missing Foundry configuration	Allow deterministic demo mode; reject Foundry strategies.
Missing Speech endpoint	Keep text modes available and explain the required endpoint and role.
Empty, oversized, unsupported, or speaker-invalid media	Reject before calling Azure Speech.
Azure Speech returns no phrases	Reject the import instead of creating an empty transcript.
Experimental deployment absent or throttled	Keep production paths unchanged and surface a Model Lab-specific error.
Calendar not approved	Do not invoke the provider action.
Session/process loss	Current in-memory work is lost unless exported.

The model calls currently have no application-level queue, retry budget, circuit breaker, or durable checkpoint. Production hardening should add bounded retries for transient errors, request-level timeouts, idempotency for external writes, structured telemetry, and clear recovery messages.

13. Observability

The application exposes user-visible warnings, evidence validation failures, latency, and model-call counts through its workflow and evaluation artifacts. Automated tests validate deterministic behavior. The deployed Container Apps environment sends runtime logs to the existing Log Analytics workspace; Application Insights remains attached to the Foundry project.

Production telemetry should include:

- Correlation ID, operation name, strategy, duration, and success status.
- Model deployment, token consumption, throttling, and retry counts.
- Connector type and failure class without transcript content or credentials.
- Validation removals and repairs as aggregate counters.
- Approval and calendar-action events with authorized reviewer identity.
- No raw transcript, prompt, model response, token, or PII in default logs.

14. Evaluation Design and Results

14.1 Fine-tuning pilot and public dataset disclosure

Production inference still uses pretrained `gpt-5.4-mini`, and `gpt-4.1-judge` remains an independent pretrained evaluation deployment. Model Lab additionally exposes the experimental `meeting-to-action-sft-v1` deployment. Its succeeded Microsoft Foundry SFT job was `ftjob-fee47a35427043ee9b4ed301464203f8`; the final model ID was `gpt-4.1-mini-2025-04-14.ft-fee47a35427043ee9b4ed301464203f8-meeting-to-action-v1`, and the deployed model ID appends `:ckpt-step-48`. User-provided website transcripts remain runtime inputs and are never accumulated into the application training dataset.

The **AMI Meeting Corpus** is the only external public dataset incorporated into the repository. The benchmark is reproducibly derived from AMI manual annotations version 1.6.2 under CC BY 4.0 from https://groups.inf.ed.ac.uk/ami/corpus/. Its 24 grounded excerpts are split into 18 development cases and six frozen test cases. The `gpt-4.1-mini` SFT pilot used 60 additional non-test AMI meetings after human-reviewed enrichment, split into 48 training and 12 validation examples. The dataset builder hard-blocks the six frozen meeting IDs, excludes all test-split records, validates exact grounding, creates a meeting-level split, and emits a provenance manifest. Generated training and review files are not committed.

Training used DeveloperTier, two epochs, a `0.5` learning-rate multiplier, service-selected batch size `1`, 366,438 trained tokens, and 336,000 billed tokens. Validation evidence favored checkpoint step 48 over the final step 96 model: deployable validation loss was `0.3387` versus `0.3543`. The best raw validation loss, `0.1937` at step 95, was not a deployable checkpoint. The frozen-set comparison used the single-model strategy for both the pretrained `gpt-4.1-mini` base and trained deployment. This promotion test isolates the effect of SFT within one model family; it is not a direct benchmark against production `gpt-5.4-mini`:

Metric	Base `gpt-4.1-mini`	SFT checkpoint	Delta
Exact entity F1	0.000	0.000	0.000
Semantic entity F1	0.000	0.054	+0.054
Deterministic grounding	1.000	1.000	0.000
Judge correctness	3.333	3.500	+0.167
Judge completeness	2.000	2.667	+0.667
Judge grounding	3.167	3.500	+0.333
Judge average	2.833	3.223	+0.390
Mean latency	11.284 s	17.785 s	+6.501 s

The quality gain with unchanged deterministic grounding justified experimental Model Lab access. The latency regression and six-case sample size do not justify replacing the production model. Separately, Model Lab performs an operational side-by-side comparison between production `gpt-5.4-mini` and the SFT deployment on a user-supplied transcript, without treating one example as benchmark evidence.

The committed six-case production baseline for `gpt-5.4-mini` reports exact F1 `0.0000`, semantic F1 `0.0289`, grounding `1.0000`, judge average `4.1683`, and mean latency `10.8378 s`. It is a separate artifact from the architecture benchmark and SFT promotion comparison.

The semantic F1 calculation uses pretrained `sentence-transformers/all-MiniLM-L6-v2` embeddings; the project neither trains that model nor includes its training corpus. Synthetic test fixtures and manually entered demonstration transcripts are not external public datasets.

14.2 Evaluation benchmark

The benchmark contains 24 annotation-linked AMI Meeting Corpus excerpts: 18 development cases and six frozen test cases. Every test case is evaluated with all three strategies. Metrics include exact entity F1, semantic entity F1, deterministic grounding rate, latency, generation calls, and an independent structured judge score.

Strategy	Exact F1	Semantic F1	Grounding	Judge average	Mean latency	Calls/case
Single model	0.000	0.057	1.000	4.000	11.588 s	1
Extractor + verifier	0.000	0.077	1.000	4.168	20.932 s	2
Specialists + consolidator	0.000	0.045	1.000	4.278	32.995 s	4

Two blinded reviewers completed 36 ratings for the 18 candidates. Human calibration passed:

- Judge-human overall MAE: 0.630, below the 0.750 maximum.
- Scores within one point of the human mean: 92.6%, above the 80% minimum.
- Inter-reviewer MAE: 0.111.
- Four of 54 candidate-dimension comparisons differed by more than one point.

The metrics disagree in meaningful ways. Exact and semantic matching are low while deterministic grounding and judge scores are high. The judge also overestimated completeness on three outlier dimensions and produced one grounding false negative. Consequently, no single metric is treated as conclusive, and the judge does not replace deterministic validation or human review.

15. Key Design Decisions

Decision	Rationale	Tradeoff
Modular monolith	Fast delivery and simple local debugging for one workflow.	No independent scaling or fault isolation between components.
Microsoft Agent Framework and Foundry	Typed agent responses and consistent orchestration across strategies.	Cloud dependency, model latency, quota, and cost.
Pydantic contracts	Machine-checkable boundaries shared by extraction, review, chat, and evaluation.	Schema evolution requires coordinated changes.
Exact evidence validation	Provides auditable support independent of model confidence.	Valid paraphrased evidence cannot replace an exact quote.
BM25 retrieval	Transparent and inexpensive for one transcript.	Lower paraphrase recall than hybrid semantic retrieval.
Extractor + verifier default	Explicit audit stage and best frozen semantic F1 at moderate	Slower and more expensive than the single model.

	call cost.	
Session-state storage	Minimal infrastructure for a capstone workflow.	No durable history, concurrency control, or multi-user audit.
Human calendar gate	Prevents unintended external actions.	Requires manual completion of each action.

16. Scalability and Evolution

The current bottleneck is model inference rather than local parsing. Specialist extraction already runs three independent calls concurrently, but a single Streamlit process still owns workflow state. For higher volume:

22. Separate the UI from an authenticated API.
23. Move extraction and evaluation to asynchronous workers with durable job state.
24. Store transcripts, approved records, and audits in tenant-partitioned persistence.
25. Add per-tenant quotas, backpressure, cancellation, and idempotency keys.
26. Introduce hybrid search only for a measured cross-meeting retrieval requirement.
27. Keep deterministic validation synchronous at the trust boundary before persistence.

17. Testing Strategy

The 70-test suite covers contracts, transcript normalization, ingestion formats, evidence validation and repair, extraction strategies, BM25 retrieval, grounded answers, review reconstruction, change auditing, calendar approval, connectors, independent-judge safeguards, semantic metrics, human calibration reporting, Speech media validation and segment conversion, supporting-context parsing, quality/conflict findings, correction events, timeline mapping, Outlook/Teams payloads, and advanced speech/conflict evaluation metrics.

Additional production tests should cover load and soak behavior, identity and tenant isolation, provider sandbox integration, accessibility, disaster recovery, prompt-injection resilience, PII redaction, and end-to-end deployment verification.

18. Known Risks and Mitigations

Risk	Impact	Current mitigation	Required production follow-up
Transcript contains sensitive data	Privacy or compliance exposure	Credentials excluded from prompts; in-memory session state	Classification, consent, redaction, retention, encryption, and DLP controls
Public unauthenticated endpoint	Unauthorized transcript submission and model-quota use	Public-access risk explicitly accepted for the capstone	Entra authentication, authorization, rate limits, and abuse monitoring
Hallucinated meeting record	Incorrect follow-up	Typed output, exact evidence validation, warnings, human	Ongoing benchmark monitoring and

		approval	escalation policy
Judge bias or inconsistency	Misleading architecture choice	Separate deployment and human calibration	Track outliers and periodically recalibrate
Session-state loss	Lost work and audit	User exports	Durable authenticated persistence and backup
Provider outage or throttling	Failed imports or model calls	Errors surfaced to user	Timeouts, retries, circuit breaking, and asynchronous jobs
Speech recognition or diarization error	Incorrect transcript attribution or timing	Human review, transcript authority, deterministic advanced metrics	Annotated multilingual speech benchmark and correction UX
Developer-tier SFT unavailable or throttled	Model Lab comparison fails	Production model remains isolated and unaffected	Availability monitoring or a supported non-production tier with explicit budget
Weak lexical retrieval	Missed relevant evidence	Approved record plus neighboring BM25 context	Retrieval evaluation and tenant-filtered hybrid search if justified
Over-privileged connector application	Excess data exposure	Read-only and delegated scopes where possible	Formal least-privilege review and conditional access

19. Open Decisions

- Decide when to replace approved public capstone access with Microsoft Entra authentication.
- Select durable stores for transcripts, approved records, and immutable audit events.
- Define transcript retention, deletion, and legal-hold requirements.
- Decide whether PII must be redacted before model processing.
- Establish service-level objectives and recovery objectives.
- Define model cost budgets and per-tenant rate limits.
- Determine when cross-meeting hybrid retrieval is justified by measured user needs.

20. Acceptance Summary

The implemented system satisfies the capstone focus areas:

- **Problem specificity:** one transcript to an approved, grounded follow-up record.
- **Data processing:** multi-format and remote-source ingestion with normalization and provenance.
- **Retrieval:** transparent BM25 selection with full-transcript citation validation.
- **Orchestration:** three measured single- and multi-agent designs with typed outputs.
- **Evaluation:** deterministic metrics, an independent LLM judge, and passed human calibration.
- **Decision reasoning:** architecture choices are tied to reliability, explainability, cost, and

measured tradeoffs.

- **Advanced meeting workflow:** media and live-clip transcription, context-aware quality review,

timestamped evidence, correction history, and approval-gated Outlook/Teams drafts.

The design is complete for a capstone and is deployed as a public Azure demonstration. Durable persistence, authenticated multi-user access, formal PII controls, stronger production observability, and resilience remain explicit requirements before enterprise use.